

Rethinking Implicit Hate Speech Detection: Focusing on Latent Hate Components via Dual-Process Argumentation

Shiqi Sun*

State Key Laboratory of AI
Safety, Institute of
Computing Technology,
Chinese Academy of
Sciences
Beijing, China
University of Chinese
Academy of Sciences
Beijing, China
sunshiqi24b@ict.ac.cn

Du Su*

State Key Laboratory of AI
Safety, Institute of
Computing Technology,
Chinese Academy of
Sciences
Beijing, China
sudu@ict.ac.cn

Wei Chen[†]

State Key Laboratory of AI
Safety, Institute of
Computing Technology,
Chinese Academy of
Sciences
Beijing, China
chenwei2022@ict.ac.cn

Xueqi Cheng

State Key Laboratory of AI
Safety, Institute of
Computing Technology,
Chinese Academy of
Sciences
Beijing, China
cxq@ict.ac.cn

Abstract

Implicit hate speech often hides harmful intent behind innocuous wording, metaphors, or hostile tone, making it difficult for detectors that rely on surface cues. We observe that large language models (LLMs) frequently exhibit pseudo-reasoning that shows over-sensitivity to spurious cues while missing the latent semantic units that actually realize hateful intent. We call these units Latent Hate Components (LHCs) and argue they should be the anchors of inference. We propose DuPL, a Dual-Process argumentation framework that centers detection on LHCs. DuPL separates (i) Mining of LHCs via a high-recall Critical Miner and a Confusion Judge that early-exits on clear cases, from (ii) Deliberation of LHCs via component-wise argumentation and a final Integrative Decision. Across IHC, SBIC, and ToxiGen, DuPL consistently outperforms prompt-learning baselines, improving accuracy by +8.36, +4.93, and +3.78 percentage points and macro-F1 by +7.18, +5.00, and +3.99 percentage points, respectively. DuPL also lowers both false positive rate and false negative rate in most settings, indicating balanced mitigation of the two failure modes common in LLM detectors. By explicitly mining and deliberating over LHCs, DuPL turns opaque, uncontrolled reasoning into structured argumentation, yielding more accurate and interpretable decisions for web moderation.

CCS Concepts

• Computing methodologies → Multi-agent systems.

Keywords

Implicit Hate Speech Detection; Large Language Models; Latent Hate Components; Multi-Agent Debate; Structured Argumentation

*Equal contribution.

[†]Corresponding author.



This work is licensed under a Creative Commons Attribution 4.0 International License. WWW '26, Dubai, United Arab Emirates.

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2307-0/2026/04

<https://doi.org/10.1145/3774904.3792159>

ACM Reference Format:

Shiqi Sun, Du Su, Wei Chen, and Xueqi Cheng. 2026. Rethinking Implicit Hate Speech Detection: Focusing on Latent Hate Components via Dual-Process Argumentation. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792159>

1 Introduction

Implicit Hate speech on the Web, particularly on online platforms and social media, directly threatens user safety and social cohesion, exacerbating polarization and marginalization [1, 28]. They often rely on subtle wording, metaphor, or hostile tone to conceal harmful intent within seemingly innocuous text components [4, 8]. These text components are termed Latent Hate Components (LHCs), on the one hand, serving as signals through which target groups can perceive hateful intent, while on the other hand acting as disguises that allow such content to evade automatic detection systems [22, 33]. Consequently, LHCs have attracted sustained attention in research on implicit hate speech [6, 15, 30].

The advent of Large Language Models (LLMs), leveraging their powerful semantic understanding and world knowledge, have presented a promising approach for deeper inference of latent hate intent [7, 21]. Recent studies have demonstrated their potential, leveraging advanced reasoning paradigms like Chain-of-Thought (CoT) and Multi-Agent Debate (MAD) to enhance detection efficacy and generalization [9, 24, 37].

Although LLMs appear to exhibit sophisticated intent understanding when detecting implicit hate speech, they in fact rely heavily on **pseudo-reasoning**. This reasoning is dominated by superficial semantics, which undermines genuine inference over latent hate components—the implicit semantic units that constitute hateful intent—which leads to two error modes tied to LHCs. Top of Figure 1 quantifies these errors under CoT and MAD paradigms. Specifically, (i) **Over-sensitivity to spurious LHCs (OSL)**: when text contains strong negative or controversial terms, the LLMs hastily accept these “decoys” and skip necessary contextual reasoning, resulting in false positives (Example 1 in bottom of Figure 1). (ii) **Omission of LHCs (OL)**: when latent hate components are hidden through implicature, metaphor, or reference, the LLMs tend

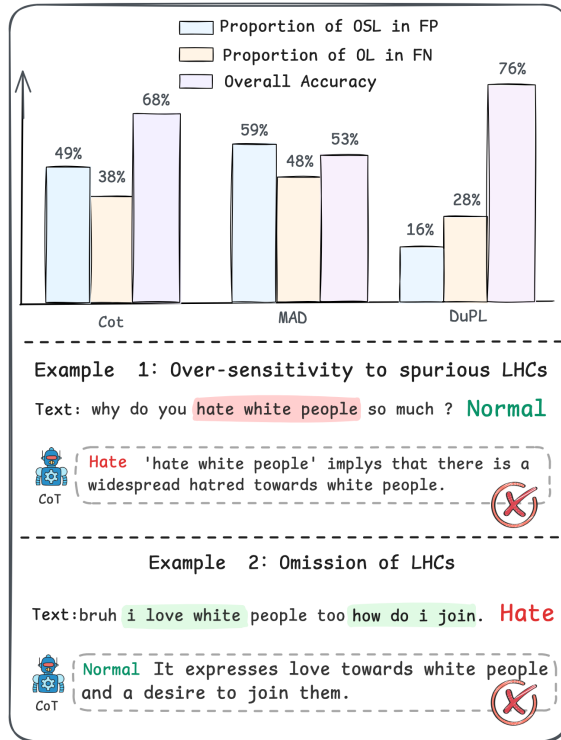


Figure 1: Overview of pseudo-reasoning errors of LLMs in implicit hate detection. Top: Under CoT and MAD (vs. DuPL), we report the proportion of Over-sensitivity to Spurious LHCs (OSL) in false positives (FP) and Omission of LHCs (OL) in false negatives (FN), together with overall accuracy. DuPL substantially reduces both error types and yields the highest accuracy. Bottom: Two representative examples — (1) “how do i join” implicates recruitment to a white-supremacist group, but the LLM adopts a literal positive reading and misses the implicated hate intent; (2) the LLM latches onto the quoted phrase “hate white people” and skips contextual reasoning, yielding a false positive.

to adopt the most straightforward, non-aggressive literal interpretation, failing to reliably uncover the underlying hateful intent and resulting in false negatives (Example 2 in bottom of Figure 1).

To overcome this pseudo-reasoning, we argue that effective detection requires a mechanism that explicitly target LHCs—ensuring broad mining of candidate latent hate components and applying deep deliberation to resolve their nuanced meanings. This dual requirement parallels human cognitive processes, where an initial fast, intuitive mechanism flags potentially problematic expressions, followed by a slower, analytical mechanism that infers their underlying intent. Motivated by these insights, we design **DuPL**, a **Dual-Process** argumentation framework centered on Latent Hate Components.

DuPL separates detection into two complementary stages. (1) **Mining of Latent Hate Components (MLHC)** stage rapidly extracts candidate components and performs preliminary judgments,

ensuring high recall while filtering trivial cases. (2) **Deliberation of Latent Hate Components (DLHC)** stage then engages multiple reasoning agents in structured debate over the retained components, and a decision aggregator consolidates their conclusions into a final verdict. By explicitly disentangling fast mining and slow deliberation, DuPL achieves a more faithful inference of latent hate intent.

DuPL emulates the human ability to make efficient yet robust decisions by coordinating these two processes, shifting the paradigm from uncontrolled, implicit reasoning to a structured argumentative process. Extensive experiments on three public datasets demonstrate that DuPL consistently outperforms state-of-the-art prompt-learning baselines: on IHC, SBIC and ToxiGen datasets, DuPL improves accuracy by +8.36 percentage points (p.p.), +4.93 p.p. and +3.78 p.p. and macro-F1 by +7.18 p.p., +5.00 p.p. and +3.99 p.p., respectively. While simultaneously reducing over-sensitivity to spurious LHCs and omission of LHCs.

In summary, our contributions can be outlined as follows:

- Reveal the critical limitations of existing prompt-learning methods in reliably detecting latent hate components.
- We introduce *latent hate components* as key semantic anchors for implicit hate speech detection, and propose DuPL—a Dual-Process argumentation framework that explicitly mines and deliberates over latent hate components.
- Extensive experiments demonstrate that DuPL consistently outperforms existing state-of-the-art prompt-learning methods, offering valuable insights for developing more transparent and robust content moderation systems.

2 Related Work

Implicit Hate Speech Detection. Early methods for implicit hate speech detection mainly relied on fine-tuning PLMs for classification and explanation [4, 10, 13, 23]. With the emergence of LLMs, recent advances increasingly leverage LLM-based reasoning approaches for implicit hate detection [8, 27]. Current research primarily focuses on harnessing different reasoning paradigms to improve both detection and explanation. Methods such as Chain-of-Thought [32] prompting and Self-reflect [12] enhance LLM performance by encouraging stepwise reasoning and critical evaluation [11]. HARE [34] further adopts CoT-style rationales to provide explanation-driven supervision. However, Zhang et al. [37] reveal that LLM-based detectors often exhibit over-sensitivity to mentions of sensitive topics, leading to frequent misclassification of benign content as hateful. PREDICT [24] introduces multi-agent debate simulations [19], where several LLMs independently reason from different perspectives before aggregating their decisions. Despite mitigating biases, this method still struggles to accurately attend to essential textual components in the reasoning process.

Multi-Agent Debate for Reasoning. Multi-Agent Debate has recently emerged as a powerful paradigm for complex reasoning tasks in NLP [5, 31]. By assigning diverse priors or stances to multiple LLM instances, MAD encourages agents to surface overlooked perspectives and engage in iterative deliberation, mitigating inherent bias in single-agent reasoning and promoting more comprehensive, factual outcomes [3, 17].

Recent research explores variants of this paradigm [2, 16, 35]. These were applied to many reasoning and alignment tasks, demonstrating consistent improvements over single-agent or self-consistency baselines [18, 19]. Importantly, multi-agent debates have also been applied to content moderation and misinformation detection [29]. This includes applications to multimodal misinformation [14], rumor verification [36], fake news detection [20], and hate speech detection [24].

3 Methodology

We instantiate DuPL as a Multi-Agent System (MAS), organized according to Dual-Process Theory. As shown in Figure 2, DuPL organizes multiple specialized agents into a two-stage pipeline: Mining of Latent Hate Components (MLHC) and Deliberation of Latent Hate Components (DLHC). All prompt templates of agents are provided in the Appendix C.

Problem Formulation. Given an input text $x = [t_1, t_2, \dots, t_n]$ consisting of n tokens, our goal is to predict a label $y \in \{\text{Hate}, \text{Normal}\}$ indicating whether x is hate speech. Implicit hate is conveyed through one or more LHCs. We define a LHC c as a span of tokens $[t_i, \dots, t_j]$ where $1 \leq i \leq j \leq n$.

Let S denote the set of all possible spans in x . Therefore, this task involves two sub-problems:

- (1) **Mining:** Critical Miner proposes candidates $\hat{C}$ and filters them to $C \subseteq \hat{C}$, then Confusion Judge issues a high-confidence judgment g based on (x, C) .
- (2) **Deliberation:** Component-wise Argumentation debates whether it actually conveys hateful intent or is *Normal* in context for each candidate $c \in C$, and then Integrative Decision aggregates these assessments into a final decision y .

3.1 Mining of Latent Hate Components

DuPL treats Latent Hate Components as semantic anchors that ground the detection of implicit hate speech. The MLHC stage implements a System 1 pathway: it executes rapid, heuristic screening centered on LHCs to resolve salient, unambiguous cases with minimal computation, and flag ambiguous texts that warrant DLHC stage deliberation. Concretely, MLHC comprises: (1) *Critical Miner* that proposes candidate Latent Hate Components; and (2) *Confusion Judge* that provides a preliminary classification decision

Critical Miner. Critical Miner is a unified, high-recall, high-confidence filtering module for rapidly and effectively mining latent hate components. It comprises two tightly coupled agents: a Candidate Miner that maximizes recall by aggressively surfacing candidates, and a Critical Challenger that promptly subjects these candidates to rigorous scrutiny, balancing coverage with reliability.

- **Candidate Miner Agent:** The recall-oriented agent scans x and proposes candidate components $\hat{C} = \text{Miner}(x)$, tolerating redundancy to maximize recall.
- **Critical Challenger Agent:** The Agent challenges the validity of each candidate component c^i in the candidate set $\hat{C}$ removing candidates that are plainly redundant or context-irrelevant, thereby yielding a refined subset $C \subseteq \hat{C}$.

A key efficiency mechanism is an Early-Exiting: If either the Candidate Miner Agent proposes no candidates ($\hat{C} = \emptyset$) or the Critical Challenger Agent prunes all candidates ($C = \emptyset$), DuPL halts and immediately returns *Normal*, indicating no credible hateful signal.

Confusion Judge. DuPL employs the Confusion Judge agent to perform a preliminary classification of the refined subset C , acting as the “gatekeeper” between MLHC and DLHC. The Confusion Judge agent produces $g = \text{Confusion Judge}(x, C)$, where $g \in \{\text{Normal}, \text{Hate}, \text{Ambiguous}\}$. If the Confusion Judge agent is confident and outputs $g \in \{\text{Normal}, \text{Hate}\}$, the system takes the decision directly and executes Early-Exiting. The *Ambiguous* label is an explicit signal that the fast path has failed and the deliberative DLHC stage should be invoked.

3.2 Deliberation of Latent Hate Components

Aligned with System 2 of Dual-Process Theory, DLHC conducts slow, deliberative analysis over the candidate Latent Hate Components. This involves (1) *Component-wise Argumentation*, where agents debate each latent hate component’s intent, and (2) *Integrative Decision*, which synthesizes insights across all components for an integrative decision.

Component-wise Argumentation. For each candidate latent hate component $c_k \in C$, DuPL constructs a specialized debate involving two autonomous agents with complementary roles:

- **Proponent Agent:** Generates arguments A_k^+ supporting that c_k is hateful within context.
- **Opponent Agent:** Generates arguments A_k^- supporting that c_k is normal within context.

In each component-level round, we fix the turn order: the Proponent speaks first and the Opponent responds. The Proponent treats the latent hate component c_k as the semantic anchor and, under both its local window and the global context x , produces A_k^+ . It explains how c_k may trigger a derogatory or exclusionary meaning, whether it interacts with other components $\{c_k\}$, and provides sentence-level textual evidence. The Opponent then produces A_k^- in the same context, offering benign readings and counter-evidence and checks consistency with speaker intent, discourse flow, and background knowledge.

Integrative Decision. Component-wise Argumentation provides local evidence of LHCs, but implicit hate can arise from the combined effect of multiple LHCs that jointly convey hateful intent. Therefore, DuPL employs the Integrative Decision agent that integrates arguments from all latent hate components into a coherent and holistic decision-making process. Specifically, DuPL concatenates all generated arguments from the Proponent agents across all components into a unified proponent perspective $\mathcal{A}^+ = \text{Concat}(A_1^+, A_2^+, \dots, A_{|C|}^+)$, and similarly concatenates arguments from the Opponent agents into a unified opponent perspective $\mathcal{A}^- = \text{Concat}(A_1^-, A_2^-, \dots, A_{|C|}^-)$. The Integrative Decision agent then conducts a comprehensive evaluation of these aggregated perspectives to reach a final integrative judgment:

$$y = \mathcal{J}(x, \mathcal{A}^+, \mathcal{A}^-) \quad (1)$$

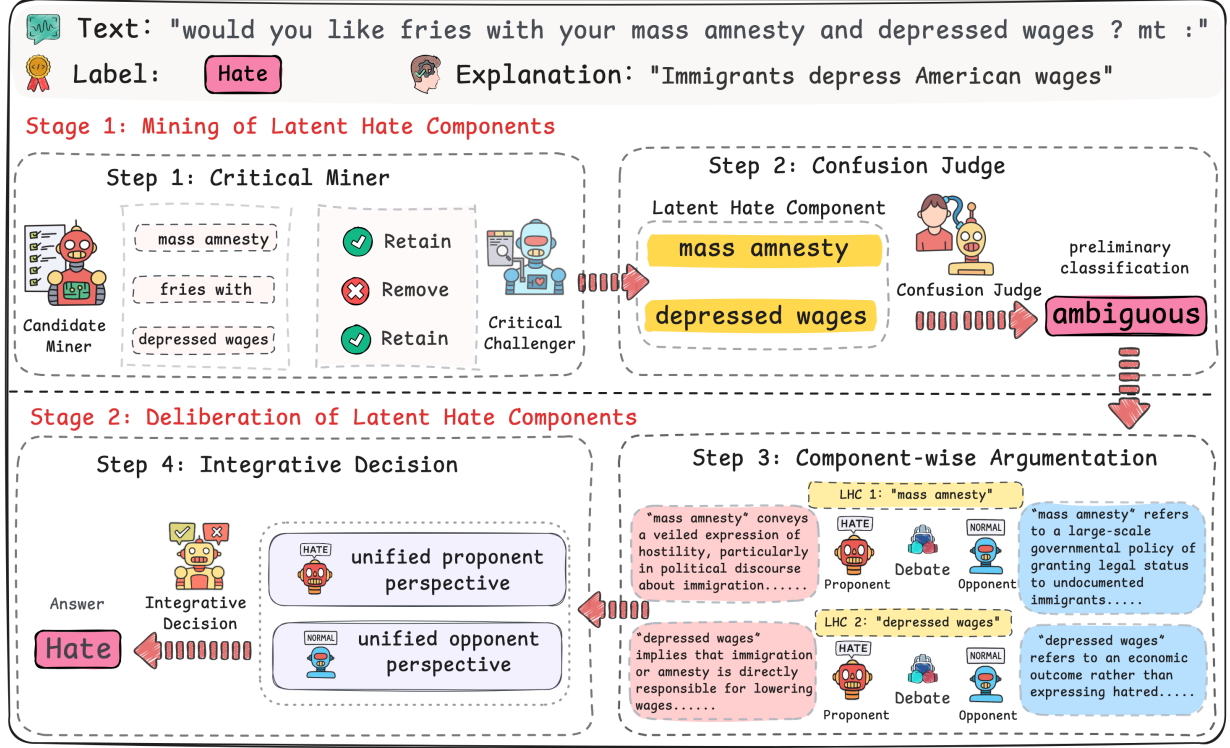


Figure 2: Overview of the DuPL framework for implicit hate speech detection.

Here, $\mathcal{J}$ denotes a balanced decision-making process that reconciles adversarial evidence through cross-comparison, integration, and disambiguation, yielding the most warranted conclusion.

4 Experiment

4.1 Experiment Setup

The primary experiments used gpt-4o-mini¹ as the backbone model. Additionally, we evaluated DuPL across two different LLMs (Qwen2.5-7B-Instruct², DeepSeek V3³) to assess its generalization ability across model types. For all experiments involving LLMs accessed through APIs, we set the following decoding configuration to ensure stable and reproducible evaluation across models: *temperature* = 0.1, *top_p* = 1.0, *n* = 1, *max_tokens* = 2048, *presence_penalty* = 0.0, and *frequency_penalty* = 0.0. For comparison methods, we used the publicly available implementations and retained all hyper-parameters and prompt templates as reported in the original papers, ensuring a fair and reproducible comparison.

We report Accuracy (Acc.) and Macro-averaged F1 scores to evaluate overall detection performance, and use False Positive Rate (FPR) and False Negative Rate (FNR) to specifically assess *Over-sensitivity to spurious LHCs* and *Omission of LHCs*, respectively. To ensure reproducibility of our results, all experiments were conducted with a fixed random seed (42), and the results were averaged over five independent runs for each configuration.

¹<https://platform.openai.com/docs/models>

²<https://www.modelscope.cn/models/qwen/Qwen2.5-7B-Instruct>

³<https://huggingface.co/deepseek-ai/DeepSeek-V3>

4.2 Main Result

Table 1 presents a comparative evaluation of our proposed DuPL framework against existing zero-shot/few-shot methods across the IHC, SBIC, and ToxiGen datasets. The results demonstrate that DuPL achieves the best overall performance across all three datasets and all three backbones. Specifically, compared to the strongest baseline method, DuPL achieves substantial improvements of 8.36 percentage point (p.p.), 4.93 p.p., and 3.78 p.p. in accuracy, and 7.18 p.p., 5.00 p.p., and 3.99 p.p. in F1 score on IHC, SBIC, and ToxiGen datasets, respectively. This consistent performance gain across diverse datasets suggests that DuPL effectively captures and reasons over latent hate components regardless of dataset-specific characteristics, demonstrating strong generalization capability. Notably, the improvement is most pronounced on the IHC dataset, which contains more complex and nuanced forms of implicit hate speech that require sophisticated contextual understanding and inference.

Compared to the strongest baseline method, DuPL reduces FPR by 11.96 p.p. and 5.80 p.p. on IHC and SBIC datasets, respectively, indicating substantial improvement in distinguishing benign content related to sensitive topics from implicit hate speech. Simultaneously, DuPL reduces FNR by 3.03 p.p. and 3.88 p.p. on these datasets, demonstrating enhanced ability to identify subtle implicit hate cues. This bidirectional improvement validates our core hypothesis that explicitly mining and deliberation latent hate components can simultaneously address both error types without trading off one for the other, unlike previous methods that often improve one metric at the expense of the other.

Table 1: Performance comparison of DuPL against baseline methods across three benchmark datasets. In addition to accuracy and F1 scores, we report False Positive Rate (FPR) and False Negative Rate (FNR) to evaluate Over-sensitivity to spurious LHCs and Omission of LHCs, respectively. Best results are highlighted in bold, and the second-best results are underlined. We report the mean $\pm$ standard deviation over the five independent repeated runs.

Method	IHC				SBIC				ToxiGen			
	Acc.(%)	F1	FPR(%)↓	FNR(%)↓	Acc.(%)	F1	FPR(%)↓	FNR(%)↓	Acc.(%)	F1	FPR(%)↓	FNR(%)↓
GPT-4o-mini												
Direct Judge	67.72 $\pm$ 2.46	67.01 $\pm$ 1.85	39.13 $\pm$ 1.52	27.34 $\pm$ 2.14	<u>80.98$\pm$3.31</u>	<u>80.79$\pm$2.55</u>	<u>13.34$\pm$2.41</u>	26.41 $\pm$ 2.40	70.08 $\pm$ 2.81	69.79 $\pm$ 2.20	22.34 $\pm$ 2.18	<u>32.41$\pm$2.44</u>
CoT[32]	<u>67.84$\pm$2.14</u>	<u>67.43$\pm$1.85</u>	38.91 $\pm$ 1.53	<u>27.13$\pm$2.45</u>	80.74 $\pm$ 2.33	80.53 $\pm$ 2.17	14.22 $\pm$ 3.14	25.23 $\pm$ 1.58	69.94 $\pm$ 2.26	69.42 $\pm$ 2.63	9.31$\pm$2.42	52.29 $\pm$ 2.51
Self-Reflect[12]	65.83 $\pm$ 2.79	65.34 $\pm$ 2.01	<u>35.31$\pm$2.42</u>	33.75 $\pm$ 2.58	79.54 $\pm$ 1.62	79.23 $\pm$ 2.23	15.31 $\pm$ 2.00	27.02 $\pm$ 1.61	<u>70.42$\pm$1.69</u>	<u>69.90$\pm$1.33</u>	13.88 $\pm$ 2.31	34.62 $\pm$ 1.42
MAD [19]	54.13 $\pm$ 2.10	53.71 $\pm$ 1.95	49.55 $\pm$ 1.80	32.53 $\pm$ 2.15	80.21 $\pm$ 2.40	80.02 $\pm$ 2.20	13.51 $\pm$ 2.12	27.18 $\pm$ 1.85	<u>67.97$\pm$2.35</u>	67.35 $\pm$ 2.14	26.73 $\pm$ 2.49	37.84 $\pm$ 2.51
PREDICT[24]	62.46 $\pm$ 2.05	58.43 $\pm$ 2.18	36.54 $\pm$ 1.75	38.14 $\pm$ 2.33	80.31 $\pm$ 2.16	80.11 $\pm$ 1.95	16.42 $\pm$ 2.01	<u>24.24$\pm$1.70</u>	69.85 $\pm$ 2.49	68.18 $\pm$ 2.20	16.05 $\pm$ 1.90	39.47 $\pm$ 2.44
DuPL	76.20$\pm$2.29	74.61$\pm$2.27	23.35$\pm$1.80	24.11$\pm$2.53	85.91$\pm$2.10	85.79$\pm$2.00	7.54$\pm$1.79	20.36$\pm$1.90	74.20$\pm$2.76	73.89$\pm$2.63	<u>12.88$\pm$2.37</u>	28.35$\pm$2.73
Qwen2.5-7B-Instruct												
Direct Judge	57.33 $\pm$ 2.18	57.33 $\pm$ 1.68	56.70 $\pm$ 1.54	14.01$\pm$1.44	70.27 $\pm$ 1.67	<u>69.94$\pm$1.23</u>	33.42 $\pm$ 1.53	27.12$\pm$0.95	62.91 $\pm$ 2.40	62.24 $\pm$ 2.00	11.40 $\pm$ 2.00	59.12 $\pm$ 2.80
CoT[32]	66.84 $\pm$ 1.25	63.42 $\pm$ 0.89	<u>29.13$\pm$1.06</u>	42.28 $\pm$ 1.14	66.18 $\pm$ 1.42	65.76 $\pm$ 1.52	39.85 $\pm$ 1.64	36.87 $\pm$ 1.21	67.17 $\pm$ 2.21	65.55 $\pm$ 1.84	<u>7.30$\pm$1.75</u>	56.80 $\pm$ 1.69
Self-Reflect [12]	61.24 $\pm$ 2.10	60.52 $\pm$ 1.95	41.08 $\pm$ 1.80	31.72 $\pm$ 2.30	62.36 $\pm$ 1.90	61.97 $\pm$ 1.75	40.15 $\pm$ 1.90	42.71 $\pm$ 1.60	65.18 $\pm$ 2.30	64.66 $\pm$ 2.10	10.45 $\pm$ 1.80	58.90 $\pm$ 2.70
MAD[19]	59.80 $\pm$ 2.00	59.20 $\pm$ 1.85	44.90 $\pm$ 1.70	<u>28.15$\pm$2.10</u>	59.42 $\pm$ 2.10	56.10 $\pm$ 1.90	45.80 $\pm$ 1.85	39.30 $\pm$ 1.70	64.22 $\pm$ 2.20	63.80 $\pm$ 2.00	12.10 $\pm$ 1.90	57.40 $\pm$ 2.60
PREDICT[24]	64.95 $\pm$ 1.90	61.10 $\pm$ 2.00	33.85 $\pm$ 1.60	35.50 $\pm$ 2.20	<u>70.33$\pm$1.95</u>	69.32 $\pm$ 1.80	33.20 $\pm$ 1.80	<u>28.05$\pm$1.60</u>	66.78 $\pm$ 2.10	65.90 $\pm$ 2.00	8.60 $\pm$ 1.70	<u>55.10$\pm$2.50</u>
DuPL	71.57$\pm$1.37	65.45$\pm$3.37	10.12$\pm$3.37	50.23 $\pm$ 3.61	71.41$\pm$3.82	70.94$\pm$4.20	10.42$\pm$4.53	45.42 $\pm$ 3.93	70.57$\pm$4.53	69.73$\pm$3.83	7.12$\pm$3.61	44.64$\pm$4.18
DeepSeek-V3												
Direct Judge	51.73 $\pm$ 1.42	51.04 $\pm$ 1.06	70.33 $\pm$ 0.93	3.04$\pm$0.93	79.68 $\pm$ 2.33	79.32 $\pm$ 2.18	34.52 $\pm$ 3.11	6.91$\pm$1.82	70.12 $\pm$ 4.23	69.15 $\pm$ 3.68	14.46 $\pm$ 3.11	35.54 $\pm$ 4.82
CoT[32]	<u>68.52$\pm$0.89</u>	<u>65.25$\pm$1.11</u>	27.45$\pm$0.84	40.61 $\pm$ 0.78	<u>81.33$\pm$2.31</u>	<u>81.19$\pm$2.44</u>	26.82 $\pm$ 2.36	14.82 $\pm$ 3.13	70.42 $\pm$ 2.60	70.18 $\pm$ 2.51	14.34 $\pm$ 2.12	33.65 $\pm$ 2.24
Self-Reflect [12]	57.40 $\pm$ 1.90	56.75 $\pm$ 1.82	49.20 $\pm$ 1.60	25.10 $\pm$ 2.05	<u>78.20$\pm$2.16</u>	<u>77.80$\pm$2.01</u>	36.40 $\pm$ 2.26	<u>8.95$\pm$1.40</u>	68.05 $\pm$ 2.31	67.30 $\pm$ 2.10	18.90 $\pm$ 2.69	37.60 $\pm$ 2.45
MAD[19]	52.36 $\pm$ 2.45	51.51 $\pm$ 2.10	53.38 $\pm$ 1.89	37.31 $\pm$ 2.22	75.50 $\pm$ 3.12	75.17 $\pm$ 2.99	<u>16.34$\pm$2.48</u>	30.33 $\pm$ 2.78	70.00 $\pm$ 1.65	69.97 $\pm$ 0.87	35.57 $\pm$ 1.19	23.96$\pm$2.19
PREDICT[24]	63.85 $\pm$ 2.61	61.90 $\pm$ 1.92	35.40 $\pm$ 1.73	39.20 $\pm$ 2.09	80.05 $\pm$ 2.70	79.70 $\pm$ 1.24	20.50 $\pm$ 1.85	24.70 $\pm$ 1.70	<u>71.10$\pm$2.45</u>	<u>70.20$\pm$2.22</u>	9.80$\pm$1.97	44.90 $\pm$ 2.53
DuPL	74.06$\pm$2.61	72.99$\pm$2.42	<u>28.70$\pm$2.12</u>	<u>18.31$\pm$2.50</u>	86.74$\pm$1.95	86.38$\pm$2.26	7.81$\pm$2.18	18.78 $\pm$ 2.10	75.88$\pm$2.87	75.53$\pm$2.26	<u>11.48$\pm$2.43</u>	<u>29.16$\pm$2.72</u>

When examining performance across different language models, DuPL exhibits robust adaptability to model architecture and scale. While using smaller models like Qwen2.5-7B-Instruct, DuPL maintains significant performance advantages over baseline methods, achieving 4.73 p.p. improvement in accuracy and 2.03 p.p. improvement in F1 score on IHC compared to the best baseline. This pattern persists across DeepSeek-V3, where DuPL’s improvements are even more pronounced (5.54 p.p., 5.41 p.p., and 4.78 p.p. accuracy gains on IHC, SBIC, and ToxiGen). The results are particularly noteworthy when considering model degradation patterns: simple prompting methods suffer dramatic performance drops when moving from larger to smaller models (e.g., Direct Judge shows a 28.80 p.p. performance decrease), while structured reasoning approaches like DuPL exhibit significantly greater robustness (only 15.90 p.p. decrease). This demonstrates that DuPL’s explicit reasoning framework provides valuable inductive bias that enables even smaller language models to perform effective hate speech detection, reducing dependency on model scale and potentially enabling more efficient real-world deployment.

4.3 Ablation Studies

We conduct ablation studies to examine the contribution of each component in DuPL. We design six controlled variants targeting both the mining and deliberation stage of the framework.

- **w/o Mining of Latent Hate Components (w/o MLHC):** Removes the Mining of Latent Hate Components stage, the Deliberation of Latent Hate Components stage is applied directly to the entire input text.
- **w/o Critical Challenger (w/o CrCh):** Removes the Critical Challenger agent, all candidates from Candidate Miner are retained without filtering.
- **w/o Confusion Judge (w/o ConJ):** Removes the Confusion Judge agent, all samples with candidate latent hate components proceed directly to the DLHC stage without preliminary classification.
- **w/o Component-wise Argumentation (w/o ComArg):** Replaces the Component-wise Argumentation with a single-perspective CoT reasoning for each candidate latent hate component.
- **w/o Integrative Decision (w/o IntDec):** Adds a classification step after Component-wise Argumentation that uses a zero-shot LLM-based classifier to make individual predictions for each candidate latent hate component, and replaces the Integrative Decision with Majority Voting over component-wise classification.

- **w/ Joint Debate (w/ JoDe):** Conducts a joint multi-agent debate over all candidate latent hate components simultaneously, instead of independent debates per latent hate component.

Table 2: Ablation study results showing the contribution of each module in DuPL.

Variant	IHC		SBIC	
	Acc.(%)	F1	Acc.(%)	F1
DuPL	76.20	74.61	85.91	85.79
<i>Mining of Latent Hate Components</i>				
w/o MLHC	71.65	71.24	80.42	80.28
w/o CrCh	70.80	70.52	76.80	73.53
w/o ConJ	73.34	72.79	82.42	82.10
<i>Deliberation of Latent Hate Components</i>				
w/o ComArg	65.42	64.15	76.95	76.53
w/o IntDec	71.50	70.75	77.24	76.97
w/ JoDe	73.43	73.20	79.61	76.44

Table 2 reports the results of six ablation variants targeting different steps of the DuPL framework. w/o MLHC causes a notable drop in accuracy (-4.55 p.p. on IHC and -5.49 p.p. on SBIC), highlighting the importance of isolating candidate spans for focused reasoning. w/o CrCh also significantly reduces performance (-5.40 p.p. on IHC and -9.11 p.p. on SBIC), suggesting that unchecked components introduce considerable noise, making benign or context-irrelevant spans pass the MLHC stage, encouraging shortcutting on surface cues and thus inflating Over-sensitivity to spurious LHCs; w/o ConJ yields a milder yet consistent decline in accuracy (-2.86 p.p. on IHC and -3.49 p.p. on SBIC), indicating that preliminary triage by the Confusion Judge helps curb false positives and prevents unnecessary deliberation on benign or easily normal cases.

In the DLHC module, w/o ComArg results in a significant F1 drop (-10.46 p.p. on IHC and -9.26 p.p. on SBIC). w/o ComArg deprives the system of adversarial scrutiny of each candidate span, making latent signals easier to miss and increasing Omission of LHCs. The w/o IntDec also leads to moderate drops (-4.70 p.p. on IHC and -8.67 p.p. on SBIC), likely because it fails to consider how different components interact in context. Finally, w/ JoDe shows smaller but still noteworthy decreases in accuracy (-2.77 p.p. on IHC and -6.30 p.p. on SBIC). These results show that each part of the framework plays a distinct role, and the overall performance depends on their coordinated interaction rather than on any single component alone.

5 Analysis

5.1 Mitigation of pseudo-reasoning

We operationalize the two pseudo-reasoning failures—Oversensitivity to spurious LHCs and Omission of LHCs—and evaluate whether DuPL mitigates them. Concretely, OSL is measured as the FPR on normal statements mentioning protected groups from Hate-Check [25], while OL is measured as the FNR on IHC samples annotated with fine-grained implicit hate types. (See Appendix A for data processing.)

As shown in Figure 3, DuPL simultaneously reduces OSL and OL without trading one for the other. In Figure 3a, DuPL yields consistently lower FPR across all protected groups—including those that typically induce high false alarms (e.g., *Black, Muslim, gay*)—indicating improved robustness to spurious, surface-level cues. Figure 3b further shows DuPL lowers FNR across all implicit hate categories on IHC, suggesting stronger recovery of subtle, context-bound LHCs.

We attribute these balanced gains to DuPL’s LHC-centric dual process. First, the MLHC stage prunes decoy components early, weakening the shortcut that causes OSL. Second, the DLHC stage enforces component-wise, adversarial argumentation before an integrative decision, making latent, non-literal signals harder to overlook, thereby reducing OL. Our ablations corroborate this mechanism: removing the Critical Challenger or Component-wise Argumentation substantially harms accuracy/F1, reflecting increased false alarms and missed implicit cues when LHCs are not filtered or deliberated explicitly (see §4.3).

5.2 Effectiveness of Latent Hate Component

Masking Ablation on Latent Hate Components. To assess the functional contribution of LHCs in DuPL, we perform a masking ablation. We compare model performance under three conditions: 0% masking with full LHC input, 50% partial masking by randomly dropping half of the LHCs, and 100% masking with the entire text as input and no LHCs provided.

As shown in Figure 4, performance degrades as more LHCs are masked, indicating that DuPL relies on structured intermediate representations beyond surface text. The FNR increases steadily, suggesting that the model cannot recover implicit hate cues without explicit reasoning anchors. In contrast, FPR peaks under partial masking—likely due to fragmented inference—and declines when all LHCs are removed, reflecting a conservative fallback strategy. These patterns underscore the role of LHCs in both enabling inference and stabilizing decisions under ambiguity.

Dynamics of LHC Filtering Across Processing Stages. To evaluate the mechanism of DuPL’s structured argumentation for identifying and filtering LHCs, we conduct a quantitative analysis focusing on the dynamic change of identified LHCs across three processing steps: Candidate Miner, Critical Challenger, and Confusion Judge.

To establish the LHC evaluation benchmark, we adopted a semi-automated annotation method: GPT-4o generated preliminary LHC annotations based on original explanations from the IHC dataset, followed by manual review and correction. In this process, 38% of the initial annotations were modified, ensuring final annotation quality. To verify reliability, we randomly sampled 100 examples for blind evaluation by independent assessors, achieving a 76% annotation agreement rate. We track the average number of LHCs retained in both hate and normal samples and compute the component-level recall—defined as the proportion of hate samples in which at least one gold LHC is successfully identified. Here, *gold LHC* refers to the human-verified ground-truth latent hate component that genuinely conveys hateful intent in a given text. As shown in Figure 5, DuPL effectively distinguishes between hate and normal content through progressive filtering. While the number of LHCs in hate samples remains relatively stable, normal samples undergo a substantial reduction, indicating the system’s ability to discard irrelevant or

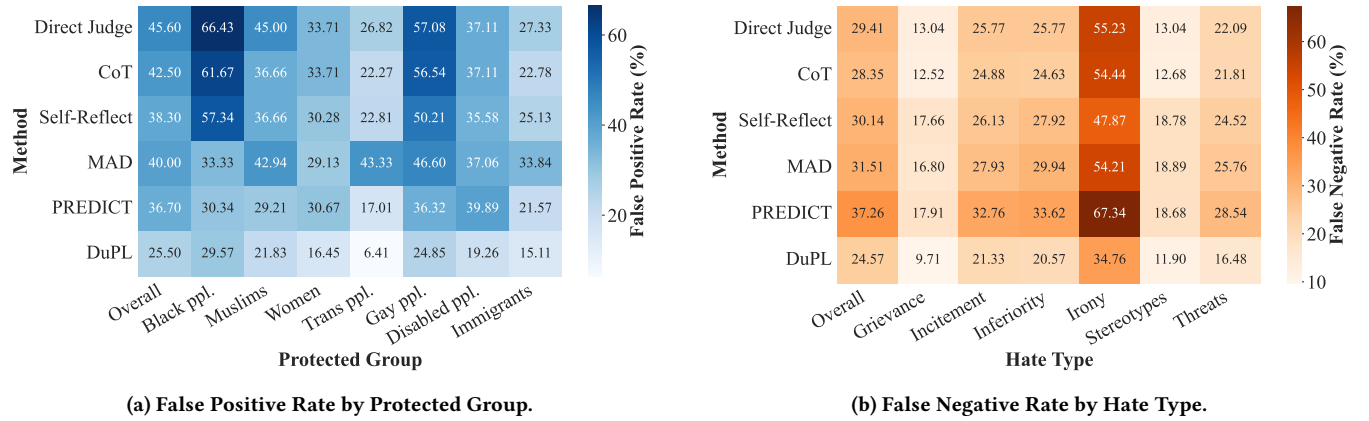


Figure 3: Comparative error analysis of DuPL and baseline methods. Sub-figure (a) plots FPR(↓) on HateCheck across seven protected groups; sub-figure (b) plots FNR(↓) on IHC across six fine-grained hate types. DuPL reduces both false positives across all protected groups and false negatives across hate categories, indicating balanced and generalizable error mitigation.

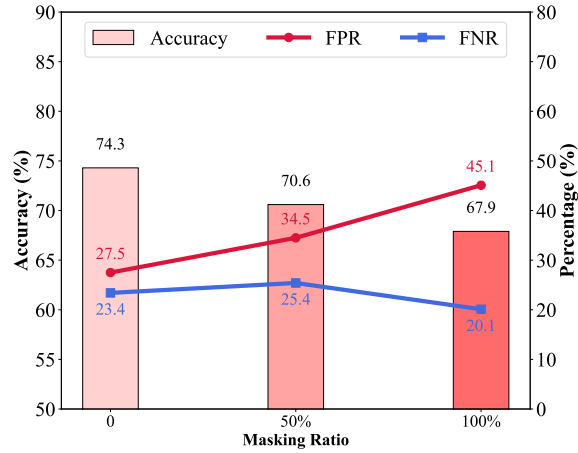


Figure 4: Impact of masking LHCs on DuPL performance. The results demonstrate that LHCs are essential for DuPL to accurately detect implicit hate, with performance dropping significantly as more LHCs are masked.

misleading components. Notably, DuPL achieves a high component-level recall of 85.8% at the Candidate Miner stage, suggesting that it captures key hate-related signals early in the reasoning process. Although recall declines in later stages, this trade-off results in a significant reduction in false positives, particularly for normal cases. These results support our hypothesis: implicit hate is often conveyed through specific textual components, and isolating these components is essential for improving both the accuracy and interpretability of hate speech detection.

5.3 Case Study

Table 4 in Appendix B demonstrates DuPL’s performance compared to existing methods on challenging cases. In case 1, the input “*trans people are a cancer*” is such a dehumanising thing to say” represents criticism of transphobic speech rather than hate speech.

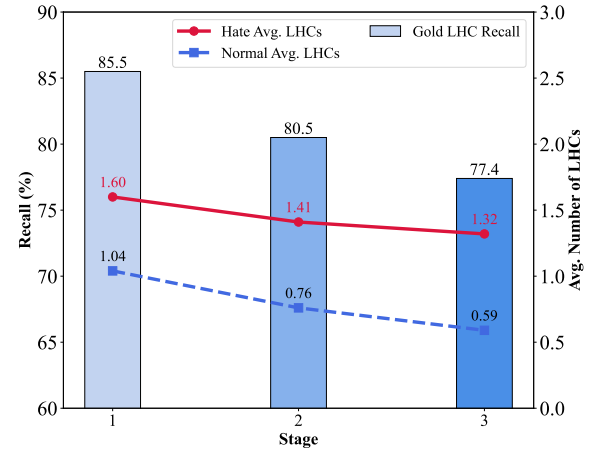


Figure 5: Average number of LHCs and component-level recall across the three stages of DuPL. The results show that DuPL effectively removes irrelevant components from normal texts while maintaining high recall of hate-related components.

CoT mistakenly focused on the isolated term “*cancer*”, interpreting it as derogatory toward transgender people. Similarly, PREDICT failed to accurately interpret the critical context, resulting in an incorrect classification. Conversely, DuPL correctly classified the text by recognizing the statement as condemning by analyzing the textual component in a broad context.

Case 2 presents “*many whites are waking up to the wonder and pride of being white. defend your race and culture. white americans are america*”, which contains implicit racial supremacist sentiments. DuPL accurately classifies “*many whites are waking up to the wonder and pride of being white...*” as “*hate*” by precisely identifying “*white americans are america*” as an exclusionary latent hate component, whereas CoT misinterprets it as merely expressing cultural pride.

The detailed processes of DuPL for this case is shown in Table 5 in Appendix B.

5.4 Computational Cost Comparison

To further evaluate the practical applicability of DuPL, we compare its computational cost against baseline methods. As shown in Table 3, DuPL achieves a balance between effectiveness and efficiency, averaging 6.3 API calls and $7.6\times$ token usage per sample, substantially lower than MAD (6.6 calls, $10.7\times$ tokens) and PREDICT (11.7 calls, $13.8\times$ tokens), yet delivering superior performance. These results highlights two insights: (i) relying solely on the MLHC stage greatly reduces token consumption and already filters out most benign cases, minimizing false positives with minimal overhead; (ii) although debate is the primary cost driver, its removal causes a significant drop in overall accuracy, confirming its critical role in resolving ambiguous cases.

Table 3: Computational cost comparison between DuPL and baseline methods. API Calls represents the average number of API calls per sample. Token Ratio shows the relative token consumption compared to Direct Judge.

Method	API Calls	Token Ratio
Direct Judge	1.0	1.0 $\times$
CoT	1.0	1.7 $\times$
Self-Reflect	2.0	2.3 $\times$
MAD	6.6	10.7 $\times$
PREDICT	11.7	13.8 $\times$
DuPL (Ours)	6.3	7.6 $\times$

Importantly, DuPL offers natural opportunities for further optimization. A dynamic Early-Exiting mechanism can be applied: When MLHC processes samples with clear intent, the DuPL directly outputs label, skipping the costly deliberation stage. As hate instances are rare in real-world moderation scenarios, this design would result in substantial cost savings and make DuPL highly advantageous in large-scale regulatory deployments.

6 Conclusion

We presented DuPL, a dual-process argumentation framework that detects implicit hate by explicitly mining and deliberating over latent hate components. By separating a fast, high-recall component mining stage from a slow, structured deliberation stage, DuPL replaces uncontrolled, opaque LLM “pseudo-reasoning” with targeted, LHC-centric inference.

Extensive experiments demonstrate that DuPL consistently outperforms prompt-learning baselines in both Accuracy and macro-F1, while simultaneously reducing over-sensitivity to spurious LHCs (false positives) and omission of LHCs (false negatives). DuPL also balances quality and cost: Early-Exiting decisions and selective deliberation reduce unnecessary debate on easy cases, making the framework amenable to large-scale moderation workflows without prohibitive compute.

We plan to (i) further optimize efficiency via adaptive debate scheduling and confidence-aware early stopping, (ii) extend DuPL to multilingual and multimodal settings where LHCs may span languages and modalities, and (iii) study human-in-the-loop protocols and dataset governance to mitigate remaining biases and enable trustworthy, transparent deployment.

Limitations

While DuPL has demonstrated effectiveness in implicit hate speech detection, limitations remain. First, what counts as “implicit hate” is socio-culturally contingent. Our definitions follow existing guidelines and thus inherit ambiguity and label noise; threshold calibration learned in one community may not transfer to others, and reclaimed slurs, quoted speech, and counter-speech remain challenging.

Second, although components provide structured rationales, they are heuristic constructs rather than causal explanations; component-level attributions should be treated as supportive evidence, not definitive proof of speaker intent. Third, performance is ultimately constrained by the capabilities of the underlying LLM (e.g., GPT-4o-mini), including model bias, outdated knowledge, and bounded reasoning ability, which can propagate through the pipeline.

Ethical Considerations

While DuPL demonstrates clear improvements in detecting implicit hate speech, we acknowledge the potential ethical risks inherent to this task. First, the subjective nature of hate speech annotation means classification boundaries are inevitably shaped by annotators’ cultural backgrounds and personal experiences. Second, our component-based detection approach may inadvertently reproduce or amplify existing biases present in the training data of LLMs, potentially leading to disparate performance across demographic groups.

To mitigate these risks, we focus on a structured argumentation approach that explicitly reasons about latent hate components. This helps reduce the risk of flagging benign content mentioning protected groups, a common source of algorithmic bias in content moderation systems.

We recognize that our method is not yet suitable for direct deployment in real-world regulatory settings, but acknowledging and addressing these risks is essential. We remain committed to strengthening protections for vulnerable communities against evolving forms of online harm. Our framework is intentionally designed to enhance transparency and explainability, advancing us toward this goal.

Acknowledgments

This work was funded by the Strategic Priority Research Program of the Chinese Academy of Sciences (Grant No. XDB0680101, No. XDB0680302), the National Key Research and Development Program of China (Grants No. 2023YFA1011602), CAS Project for Young Scientists in Basic Research (Grant No. YSBR-034), Xiaomi Young Talents Program, and Innovation Funding Programs of State Key Laboratory of AI Safety.

References

- [1] Natalia Abuín-Vences, Ubaldo Cuesta-Cambra, José-Ignacio Niño-González, and Carolina Bengochea-González. 2022. Hate speech analysis as a function of ideology: Emotional and cognitive effects. *Comunicar: Media Education Research Journal* 30, 71 (2022), 35–45.
- [2] Justin Chen, Swarnadeep Saha, and Mohit Bansal. 2024. Reconcile: Round-table conference improves reasoning via consensus among diverse llms. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 7066–7085.
- [3] Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. 2024. Improving factuality and reasoning in language models through multiagent debate. In *Proceedings of the 41st International Conference on Machine Learning*. JMLR.org, Article 467, 31 pages.
- [4] Mai ElSherief, Caleb Ziems, David Muchlinski, Vaishnavi Anupindi, Jordyn Seybolt, Munmun De Choudhury, and Diyi Yang. 2021. Latent Hatred: A Benchmark for Understanding Implicit Hate Speech. 345–363.
- [5] Yi Fang, Moxin Li, Wenjie Wang, Lin Hui, and Fuli Feng. 2025. Counterfactual debating with preset stances for hallucination elimination of LLMs. In *Proceedings of the 31st International Conference on Computational Linguistics*. 10554–10568.
- [6] Herbert Paul Grice. 1975. Logic and conversation. *Syntax and semantics* 3 (1975), 43–58.
- [7] Keyan Guo, Alexander Hu, Jaden Mu, Ziheng Shi, Ziming Zhao, Nishant Vishwamitra, and Hongxin Hu. 2023. An Investigation of Large Language Models for Real-World Hate Speech Detection. *2023 International Conference on Machine Learning and Applications (ICMLA)* (2023), 1568–1573.
- [8] Thomas Hartvigsen, Saadia Gabriel, Hamid Palangi, Maarten Sap, Dipankar Ray, and Ece Kamar. 2022. ToxiGen: A Large-Scale Machine-Generated Dataset for Adversarial and Implicit Hate Speech Detection. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 3309–3326.
- [9] Ming Shan Hee, Shivam Sharma, Rui Cao, Palash Nandi, Preslav Nakov, Tanmoy Chakraborty, and Roy Ka-Wei Lee. 2024. Recent advances in online hate speech moderation: Multimodality and the role of large models. *Findings of the Association for Computational Linguistics: EMNLP 2024* (2024), 4407–4419.
- [10] Nhat M. Hoang, Xuan Long Do, Duc Anh Do, Duc Anh Vu, and Luu Anh Tuan. 2024. ToXCL: A Unified Framework for Toxic Speech Detection and Explanation. Mexico City, Mexico, 6460–6472.
- [11] Fan Huang, Haewoon Kwak, and Jisun An. 2023. Chain of Explanation: New Prompting Method to Generate Quality Natural Language Explanation for Implicit Hate Speech. In *Companion Proceedings of the ACM Web Conference 2023*. Association for Computing Machinery, New York, NY, USA, 90–93.
- [12] Ziwei Ji, Tiezheng Yu, Yan Xu, Nayeon Lee, Etsuko Ishii, and Pascale Fung. 2023. Towards mitigating LLM hallucination via self reflection. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. 1827–1843.
- [13] Youngwook Kim, Shinwoo Park, and Yo-Sub Han. 2022. Generalizable implicit hate speech detection using contrastive learning. In *Proceedings of the 29th International Conference on Computational Linguistics*. 6667–6679.
- [14] Kumud Lakara, Georgia Channing, Juil Sock, Christian Rupprecht, Philip Torr, John Collomosse, and Christian Schroeder de Witt. 2025. LLM-Consensus: Multi-Agent Debate for Visual Misinformation Detection. arXiv:2410.20140
- [15] Yejin Lee, Joonghyuk Hahn, Hyeseon Ahn, and Yo-Sub Han. 2025. AmpleHate: Amplifying the Attention for Versatile Implicit Hate Detection. arXiv:2505.19528
- [16] Guohao Li, Hasan Abed Al Kader Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. 2023. CAMEL: communicative agents for "mind" exploration of large language model society. In *Proceedings of the 37th International Conference on Neural Information Processing Systems*. Curran Associates Inc., Article 2264, 18 pages.
- [17] Moxin Li, Wenjie Wang, Fuli Feng, Fengbin Zhu, Qifan Wang, and Tat-Seng Chua. 2024. Think Twice Before Trusting: Self-Detection for Large Language Models through Comprehensive Answer Reflection. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. 11858–11875.
- [18] Yunxuan Li, Yibing Du, Jiageng Zhang, Le Hou, Peter Grabowski, Yeqing Li, and Eugene Ie. 2024. Improving Multi-Agent Debate with Sparse Communication Topology. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. 7281–7294.
- [19] Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang, Yujia Yang, Shuming Shi, and Zhaopeng Tu. 2024. Encouraging divergent thinking in large language models through multi-agent debate. In *Proceedings of the 2024 conference on empirical methods in natural language processing*. 17889–17904.
- [20] Yuhua Liu, Yuxuan Liu, Xiaoping Zhang, Xiuying Chen, and Rui Yan. 2025. The Truth Becomes Clearer Through Debate! Multi-Agent Systems with Large Language Models Unmask Fake News. arXiv:2505.08532
- [21] Sarah Masud, Sahajpreet Singh, Viktor Hangya, Alexander Fraser, and Tanmoy Chakraborty. 2024. Hate Personified: Investigating the role of LLMs in content moderation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. 15847–15863.
- [22] Nicolás Benjamín Ocampo, Elena Cabrio, and Serena Villata. 2023. Unmasking the hidden meaning: Bridging implicit and explicit hate speech embedding representations. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. 6626–6637.
- [23] Nicolás Benjamín Ocampo, Ekaterina Sviridova, Elena Cabrio, and Serena Villata. 2023. An in-depth analysis of implicit and subtle hate speech messages. In *EACL 2023-17th Conference of the European Chapter of the Association for Computational Linguistics*, Vol. 2023. Association for Computational Linguistics, 1997–2013.
- [24] Someen Park, Jaehoon Kim, Seungwan Jin, Sohyun Park, and Kyungsik Han. 2024. PREDICT: multi-agent-based debate simulation for generalized hate speech detection. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. 20963–20987.
- [25] Paul Röttger, Bertie Vidgen, Dong Nguyen, Zeerak Talat, Helen Margetts, and Janet Pierrehumbert. 2021. HateCheck: Functional tests for hate speech detection models. In *Proceedings of the 59th annual meeting of the association for computational linguistics and the 11th international joint conference on natural language processing (volume 1: long papers)*. 41–58.
- [26] Maarten Sap, Saadia Gabriel, Lianhui Qin, Dan Jurafsky, Noah A Smith, and Yejin Choi. 2020. Social bias frames: Reasoning about social and power implications of language. In *Proceedings of the 58th annual meeting of the association for computational linguistics*. 5477–5490.
- [27] Xinyue Shen, Yixin Wu, Yiting Qu, Michael Backes, Savvas Zannettou, and Yang Zhang. 2025. HateBench: Benchmarking Hate Speech Detectors on LLM-Generated Content and Hate Campaigns. arXiv:2501.16750
- [28] Leandro Silva, Mainack Mondal, Denzil Correa, Fabricio Benevenuto, and Ingmar Weber. [n. d.]. Analyzing the Targets of Hate in Online Social Media. *Proceedings of the International AAAI Conference on Web and Social Media* 10, 1 ([n. d.]), 687–690.
- [29] Andries Smit, Nathan Grinsztajn, Paul Duckworth, Thomas D. Barrett, and Arnau Pretorius. 2024. Should we be going MAD? a look at multi-agent debate strategies for LLMs. In *Proceedings of the 41st International Conference on Machine Learning*. JMLR.org, Article 1866, 23 pages.
- [30] Dan Sperber and Deirdre Wilson. 1986. *Relevance: Communication and cognition*. Vol. 142. Harvard University Press Cambridge, MA.
- [31] Sijia Wang and Lifu Huang. 2024. Debate as Optimization: Adaptive Conformal Prediction and Diverse Retrieval for Event Extraction. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. 16422–16435.
- [32] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Proceedings of the 36th International Conference on Neural Information Processing Systems*. Curran Associates Inc., Red Hook, NY, USA, Article 1800, 14 pages.
- [33] Lu Wei, Liangzhi Li, Tong Xiang, Liu Xiao, and Noa Garcia. 2025. Cracking the Code: Enhancing Implicit Hate Speech Detection through Coding Classification. In *Proceedings of the 5th Workshop on Trustworthy NLP (TrustNLP 2025)*. 112–126.
- [34] Yongjin Yang, Joonkee Kim, Yujin Kim, Namgyu Ho, James Thorne, and Se-Young Yun. 2023. HARE: Explainable Hate Speech Detection with Step-by-Step Reasoning. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. 5490–5505.
- [35] Zhangye Yin, Qiushi Sun, Cheng Chang, Qipeng Guo, Junqi Dai, Xuan-Jing Huang, and Xipeng Qiu. 2023. Exchange-of-thought: Enhancing large language model capabilities through cross-model communication. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. 15135–15153.
- [36] Yuting Zeng, Weizhe Huang, Lei Jiang, Tongxuan Liu, XiTai Jin, Chen Tianying, Tian, Jing Li, and Xiaohua Xu. 2025. S2-MAD: Breaking the Token Barrier to Enhance Multi-Agent Debate Efficiency. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. 9393–9408.
- [37] Min Zhang, Jianfeng He, Taoran Ji, and Chang-Tien Lu. 2024. Don't Go To Extremes: Revealing the Excessive Sensitivity and Calibration Limitations of LLMs in Implicit Hate Speech Detection. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 12073–12086.

A Datasets and Baselines

Baselines. We compare DuPL against several prompt-learning methods for hate speech detection:

- **Direct Judge:** A straightforward approach where LLMs directly classify whether input text contains hate speech without additional reasoning steps.
- **CoT [32]:** Chain-of-Thought prompting that guides LLMs to generate step-by-step reasoning before making a classification decision.

Table 4: Case study examples comparing DuPL with CoT and PREDICT

Case 1
INPUT TEXT: “‘trans people are a cancer’ is such a dehumanising thing to say.” GROUND TRUTH: NORMAL Explanation of CoT: HATE “we need to analyze the statement ‘trans people are a cancer.’ This statement is highly offensive and dehumanizing towards transgender individuals. Referring to a group of people as a “cancer” is a form of hate speech that promotes discrimination and intolerance.” Explanation of PREDICT: HATE “The affirmative side effectively highlights the problematic nature of the statement by emphasizing the potential for normalization of hate speech and the implications of downplaying white supremacy. The negative side, while presenting valid points about context and freedom of expression, does not sufficiently address the risks associated with conflating cultural pride with supremacist ideologies. The affirmative argument is more compelling in recognizing the nuances of hate speech and its impact on societal discourse.” Explanation of DuPL: NORMAL “The statement condemns dehumanizing language towards transgender individuals, advocating for respect and empathy. It does not promote hate but highlights the harmful impact of such rhetoric.”
Case 2
INPUT TEXT: “many whites are waking up to the wonder and pride of being white. defend your race and culture. white americans are america” GROUND TRUTH: HATE Explanation: “White people should be proud to be white.” Explanation of CoT: NORMAL “This statement appears to be promoting pride in one’s race and culture, which is a common sentiment among many different racial and ethnic groups. It does not contain any explicit hate speech or derogatory language towards any other group.” Explanation of PREDICT: HATE “The affirmative side effectively highlights the potential implications of the statements, emphasizing how they can foster division and exclusion, which aligns with the characteristics of hate speech. The negative side, while presenting a valid perspective on cultural pride, does not adequately address the exclusionary nature of the statements in the context of racial dynamics in America.” Explanation of DuPL: HATE “Explanation for judgement as implicit hate speech: The phrase ‘defend your race and culture’ in the input text implies a sense of superiority and exclusivity based on race. By emphasizing the need to defend one’s race, it suggests a perceived threat or attack on white identity, promoting a divisive and discriminatory mindset. This language can be seen as implicit hate speech as it reinforces harmful stereotypes and promotes the idea of racial superiority.”

- **Self-Reflect** [12]: Encourages LLMs to first make a judgment, then critically reflect on that judgment before providing a final decision.
- **MAD** [19]: Multi-Agent Debate framework where multiple LLM instances debate whether content is hateful before reaching a consensus.
- **PREDICT** [24]: A multi-agent detection architecture that simulates diverse labeling perspectives through structured debates.

Datasets. To evaluate the effectiveness of DuPL, We conducted experiments on the IHC, SBIC and ToxiGen datasets.

- **Implicit Hate Speech Corpus (IHC)** [4]: The dataset focuses on implicit hate speech collected from Twitter. It comprises 19,112 tweets, with 4,909 labeled as implicit hate and 933 labeled as explicit hate.
- **Social Bias Inference Corpus (SBIC)** [26]: The dataset with hierarchical categories of social biases and stereotypes. It supports large-scale modeling and evaluation with 150k structured annotations of social media posts, covering over 34k implications about demographic groups.
- **ToxiGen** [8]: This dataset includes 274,186 machine-generated statements created by GPT-3. ToxiGen explicitly targets bias detection concerning 13 specific minority groups, including racial, religious, and LGBTQ+ communities.

In our main experiments, we evaluate our method on the full test sets of IHC, SBIC and ToxiGen datasets. These datasets were accessed through their respective repositories and were formatted to ensure compatibility with our processing pipeline.

In the analysis, we reused the same IHC hate samples. For non-hate data, we sampled 500 examples from HateCheck [25], a standard test suite containing normal statements that mention protected groups. Our subset covers seven groups: women, transgender people, gay people, disabled individuals, Black people, immigrants, and Muslims. These examples ensure relevance and neutrality, enabling controlled sensitivity evaluation.

B Examples

Table 4 and Table 5 present representative case studies, illustrating how DuPL identifies latent hate components and performs structured argumentation compared to baseline methods on challenging implicit hate speech examples.

C Prompts

Table 6 presents all the prompt templates used in the DuPL framework. Each template corresponds to a specific module or evaluation setting within DuPL, ensuring consistency and reproducibility in experimental results.

Table 5: An example of DuPL processing on implicit hate speech

Stage	Output Content
Input Text	"many whites are waking up to the wonder and pride of being white . defend your race and culture . white americans are america"
Ground Truth	HATE
Explanation	"White people should be proud to be white."
Step 1: Mining of Latent Hate Components (MLHC)	
Candidate Miner	Candidate Latent Hate Components: (1) "waking up to the wonder and pride of being white" (2) "white americans are america"
Critical Challenger	Retained Latent Hate Components : (1) "waking up to the wonder and pride of being white" (2) "white americans are america" Removed Latent Hate Components : []
Confusion Judge	Ambiguous
Step 2: Deliberation of Latent Hate Components (DLHC)	
Component-wise Argumentation	Debate for LHC (1) : "waking up to the wonder and pride of being white" Proponent: This statement implies that being white is something to be proud of and that white Americans should defend their race and culture. This can be seen as implicit hate speech because it promotes the idea of white superiority and suggests that white people need to protect their race from perceived threats...." Opponent: "Explanation of normal speech: The text can be seen as normal speech because it is expressing pride in one's race and culture, which is a common sentiment among many individuals regardless of their race. It is not explicitly promoting hate or discrimination towards other races, but rather encouraging individuals to embrace and defend their own heritage. " Judge: NORMAL Reason: " Explanation of normal speech: The text can be seen as normal speech because it is expressing pride in one's race and culture, which is a common sentiment among many individuals regardless of their race. Overall, the text is more about celebrating one's identity rather than promoting hate towards others.", " Debate for LHC (2) : "white americans are america" Proponent: "This statement implies that white Americans are the sole representation of America, excluding and marginalizing other racial and cultural groups. By emphasizing the importance of defending one's race and culture, it suggests a superiority complex and promotes the idea of white supremacy. Overall, the text promotes implicit hate speech by promoting exclusion and superiority based on race." Opponent: "The INPUT TEXT can be considered normal speech because it simply expresses pride in being white and encourages individuals to defend their race and culture. The statement white Americans are American can be interpreted as a celebration of the contributions and identity of white Americans within the broader American society, rather than an exclusionary or supremacist statement." Judge: HATE Reason: "This statement implies that white Americans are the sole representation of America, excluding and marginalizing other racial and cultural groups. This language reinforces harmful stereotypes and perpetuates division and discrimination based on race. Overall, the text promotes implicit hate speech by promoting exclusion and superiority based on race."
Integrative Decision	Final Decision: HATE Reason: "Explanation for judgement as implicit hate speech: The phrase "defend your race and culture" in the input text implies a sense of superiority and exclusivity based on race. By emphasizing the need to defend one's race, it suggests a perceived threat or attack on white identity, promoting a divisive and discriminatory mindset. This language can be seen as implicit hate speech as it reinforces harmful stereotypes and promotes the idea of racial superiority."

Table 6: Prompt templates for each processing stage.

Mining of Latent Hate Components (MLHC) Prompt of Candidate Miner: You are an expert specialized in detecting implicit hate speech. Carefully analyze the INPUT TEXT to detect any (implicit and explicit) hate, bias, or discriminatory intent hidden within seemingly neutral or subtle language. 1. Semantic Cues: Contemptuous Tone; Unequal Comparisons; Negative Labeling; Subtle Devaluation or Disguised Denigration; Incomplete or Dehumanizing Metaphors; Conditional Compliments 2. Reasoning Cues: Label and Conclusion Blurring; Falsely Neutral Logic; Biased Causality 3. Emotive Cues: Dehumanizing or Pathological Metaphors; Antagonistic Sentiments Your analysis should not rely solely on a single word or fragment, but rather on the overall text or context. - Extract the Sentence Components (continuous spans of tokens) that carry such implicit negativity toward a group. - If no abnormal Sentence Component is confidently detected, output: [] Output Format: - Each result should be a triple: ("Sentence Component", Target Group, Implied meaning/Bias/Intent) - Final Output: [] or [(triple 1), (triple 2), ...] INPUT TEXT: {input_text}
Prompt of Critical Challenger: You are a professional implicit hate speech analysis expert. Your task is to critically evaluate each proposed triple in its full context. For each triple, decide: - 'retain': if it expresses implicit hate or dehumanization toward a group. - 'remove': If it is ambiguous, oversensitive, or does not reflect hateful intent in the context. You need to avoid being overly sensitive to single words, ingredients that touch on sensitive topics, and fully consider the rationality of opposing views Use this strict output format for each triple: 'retain <SEP> <EXPLANATION>' or 'remove <SEP> <EXPLANATION>' INPUT TEXT: {input_text} Triple of Latent Hate Component: {lhc}
Prompt of Confusion Judge: You are a hate speech detection expert. Please analyze the input text and classify it into one of the following categories: - 'hate' – Only if the context is very clear and clear to convey hatred, it is directly judged as hate. - 'normal' – If the text contains no hate or harmful language and does not need further analysis, even if it is critical or negative in tone. - 'ambiguous' – If there is any potential ambiguity and uncertainty in the text, or subtle and may require further review to determine intent. INPUT TEXT: {input_text} Focus on Text Components: {lhc} These ingredients may easily lead to over-sensitivity or neglect of contextual meaning. The Output can be either of the form : "normal <SEP> <Explanation>" or "hate <SEP> <Explanation>" or "ambiguous <SEP> <Explanation>"
Deliberation of Latent Hate Components (DLHC) Prompt of Proponent Agent: You think the INPUT TEXT may be implicit hate speech. Carefully analyze the content and context to identify any elements that attack, demean, or express prejudice against a protected or vulnerable group. - Use the provided signals (Latent Hate Component) as evidence to support your argument, but interpret them in context – focus only on genuine evidence of hateful intent. - Construct a clear, logical argument explaining why the content should be labeled as hate speech. Avoid exaggeration or misinterpretation: if a signal is ambiguous, acknowledge it but clarify why it still suggests hate (if applicable). within 100 words. INPUT TEXT: {input_text} Latent Hate Component: {lhc} Prompt of Opponent Agent: You think The INPUT TEXT is normal speech, the text does not constitute implicit hate speech. Carefully examine the content and explain why it should be considered normal, emphasizing benign or context-driven interpretations. - Address provided Latent Hate Component or potential bias, and refute or contextualize them to show they lack hateful intent. - Acknowledge any negative language or tone in the content, but show that it is not directed at a protected group or not meant as a slur. within 100 words. INPUT TEXT: {input_text} Latent Hate Component: {lhc} Explanation of Proponent: {exp_hate}
Prompt of Integrative Decision: You are now an impartial judge who has heard arguments from both sides. Your task is to decide whether the content is hate speech or normal after evaluating both arguments and the content itself. Carefully consider the evidence and reasoning presented by the prosecution (hate) and the defense (normal). Determine which side's argument is more convincing and better supported by the actual content. Base your judgment on the content's features: - if the content clearly contains slurs, insults, attacks or derogatory stereotypes aimed at a group (even implicitly), then the hate classification is warranted. - If the content lacks clear evidence of group-directed hate and the hate argument relied on stretches or misinterpretation, then the content should be deemed normal. - In ambiguous or borderline cases, weigh the arguments and evidence carefully and lean toward the side with more substantial support from the content. - Do not default to calling something "hate" without strong evidence – ensure that a hate ruling is firmly justified by the content's intent. Output can be either of the form : "normal <SEP> <Explanation>" or "hate <SEP> <Explanation>". INPUT TEXT: {input_text} Explanation of Proponent: {exp_hate} Explanation of Opponent: {exp_normal}